SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME (CO2): Configure IP addressing schemes, subnetting, and basic routing techniques for efficient network design and topology.

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Answer Presentation and Calculation Method

The answers below are organized by question and task, with calculation steps, summary tables, and conclusions. For IPv4, usable host count = 2^h − 2, where h is the number of host bits. The network and broadcast addresses are excluded from ordinary host assignment.

Question 1 — Smart Campus IPv4 Subnet Allocation (10 Marks)

Given block: 192.168.100.0/24. Its full address range is 192.168.100.0–192.168.100.255 (256 total addresses). To avoid overlap, the specified subnets are allocated consecutively from the start of the /24 block.

Department	Prefix / Mask	Network Address	Usable Host Range	Broadcast Address	Usable Hosts
School of Computing	/25 255.255.255.128	192.168.100.0	192.168.100.1–192.168.100.126	192.168.100.127	$2^7 - 2 = 126$
School of Electronics	/26 255.255.255.192	192.168.100.128	192.168.100.129–192.168.100.190	192.168.100.191	$2^6 - 2 = 62$
School of Mechanical Engineering	/27 255.255.255.224	192.168.100.192	192.168.100.193–192.168.100.222	192.168.100.223	$2^5 - 2 = 30$

Remaining unused range

The allocated subnets cover addresses from 192.168.100.0 through 192.168.100.223. Therefore, the unused range is 192.168.100.224–192.168.100.255, which aligns exactly with 192.168.100.224/27. It contains 32 total addresses, or 30 usable host addresses if assigned as a subnet.

Conclusion

All three subnets are correctly aligned, do not overlap, and fit within the original /24 block.

Question 2 — Company Department Subnetting (10 Marks)

Given block: 10.10.0.0/16, covering 10.10.0.0–10.10.255.255 (65,536 total addresses). The subnets below are allocated consecutively from the start of the block, with no overlap.

Department	Prefix / Mask	Network Address	Usable Host Range	Broadcast Address	Usable Hosts
Software Development	/24 255.255.255.0	10.10.0.0	10.10.0.1–10.10.0.254	10.10.0.255	$2^8 - 2 = 254$
Quality Assurance	/25 255.255.255.128	10.10.1.0	10.10.1.1–10.10.1.126	10.10.1.127	$2^7 - 2 = 126$
Human Resources	/26 255.255.255.192	10.10.1.128	10.10.1.129–10.10.1.190	10.10.1.191	$2^6 - 2 = 62$
Executive Management	/27 255.255.255.224	10.10.1.192	10.10.1.193–10.10.1.222	10.10.1.223	$2^5 - 2 = 30$

Remaining unused range

After the four allocations, the remaining unallocated range is 10.10.1.224–10.10.255.255. This is the unused continuous address range within the original /16 block; it is not one single CIDR subnet. It can be divided into multiple valid CIDR blocks when further allocation is needed.

Conclusion

Each department receives a subnet of the requested size, and all assigned ranges remain inside 10.10.0.0/16 without overlapping.

Question 3 — IPv6 Addressing (10 Marks)

Given IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Task	Answer and Explanation
1. Compressed notation	2001:db8:abcd::/64. Leading zeros in each hextet are omitted, and the longest consecutive sequence of zero hextets is replaced by ::.
2. Expanded notation	2001:0db8:abcd:0000:0000:0000:0000:0000/64. The :: represents five zero hextets in this address.
3. Network prefix	2001:0db8:abcd:0000::/64. The first 64 bits (the first four hextets) identify the network prefix.
4–5. Total possible addresses	A /64 leaves 64 bits for the interface identifier: 2^{64} = 18,446,744,073,709,551,616 possible addresses.

Important IPv6 note: IPv6 does not use broadcast addresses, so the IPv4 rule of subtracting two addresses does not apply. The mathematical size of the /64 subnet is 2^{64} addresses; practical address-assignment rules may limit which addresses are assigned to devices.

Conclusion

The given IPv6 address compresses to 2001:db8:abcd::/64 and represents a subnet with 64 bits in the network prefix and 64 bits in the interface identifier.

Question 4 — Routing Table and Packet Forwarding (10 Marks)

Available routing entries: 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24. Destination IP address: 192.168.2.45.

Task	Answer and Reason
1. Destination subnet	192.168.2.0/24. The address 192.168.2.45 belongs to the range 192.168.2.0–192.168.2.255; ordinary host addresses are 192.168.2.1–192.168.2.254.
2. Matching route	192.168.2.0/24. The destination matches this routing-table entry.
3. Next hop / outgoing interface	Use the next-hop IP address or outgoing interface associated with the 192.168.2.0/24 route. The question gives no next-hop address or interface name, so a specific value cannot be stated. If the network is directly connected, the router sends the packet through the interface connected to that subnet.
4. Default route	0.0.0.0/0. This route is used only if configured and no more-specific route matches the destination. Its next hop or exit interface must be defined in the router configuration.

Forwarding decision

The router checks the destination IP against its routes and uses the longest-prefix match. Here, 192.168.2.0/24 is the matching route, so the packet is forwarded using that route rather than the default route.

Conclusion

The destination subnet and matching entry can be identified from the supplied table. The actual next hop depends on route configuration not included in the scenario.

Question 5 — Administration and Finance LANs (10 Marks)

Both LANs use /26, equivalent to subnet mask 255.255.255.192. A /26 has 64 total addresses and 62 usable host addresses.

LAN	Network Address	Valid Host Range	Broadcast Address	Default Gateway
Administration	192.168.10.0/26	192.168.10.1–192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65–192.168.10.126	192.168.10.127	192.168.10.65

Suggested computer addresses

LAN	Computer 1	Computer 2	Computer 3	Subnet Mask	Gateway
Administration	192.168.10.2	192.168.10.3	192.168.10.4	255.255.255.192	192.168.10.1
Finance	192.168.10.66	192.168.10.67	192.168.10.68	255.255.255.192	192.168.10.65

Administration computers should use 192.168.10.1 as the default gateway. Finance computers should use 192.168.10.65. The suggested computer addresses are within the valid host ranges and do not duplicate the gateway addresses.

Conclusion

The two /26 networks are adjacent, correctly aligned, and non-overlapping. Each LAN has its own gateway within its valid host range.

Overall Summary

- Q1 and Q2: IPv4 subnet allocations are aligned to the requested prefix lengths and avoid overlap.
- Q3: IPv6 compression and expansion are correct; a /64 contains 2^{64} possible addresses and has no broadcast address.
- Q4: The correct route is 192.168.2.0/24; a default route is 0.0.0.0/0 when configured.
- Q5: Gateway and sample host addresses are valid and distinct.

Academic integrity: Use this document as a structured study draft. Review the calculations, follow your institution's submission rules, and ensure you can explain the reasoning in your own words.